

Performance Testing

Project: Employee Software Installation Request (ServiceNow)

1. Objective

The objective of performance testing is to evaluate the responsiveness, reliability, and efficiency of the **Employee Software Installation Request** application developed on the ServiceNow platform. The testing verifies that the Service Catalog form, approval workflow, and Business Rules execute within acceptable response times under normal and moderate workloads without causing errors or performance degradation.

2. Scope of Performance Testing

The performance testing covered the following functional areas of the application:

- Loading time of the **Software Installation Request** Service Catalog form.
- Time required to submit a software installation request.
- Execution time of the **Flow Designer** approval workflow.
- Automatic creation of manager approval records.
- Processing of approval and rejection actions.
- Execution of **Business Rules** for automatic Incident creation during software license shortages.
- System behavior during multiple concurrent software request submissions.
- Email notification delivery after request approval or rejection.

3. Test Environment

The tests were performed in the following development environment.

Component	Details
Platform	ServiceNow Personal Developer Instance (PDI)
Application	Employee Software Installation Request
Test Users	Simulated Employee, Manager, and IT/Network Support user accounts
Testing Tools	Flow Designer Execution Details, System Logs, Manual Response Time Measurement
Test Data	Multiple software installation requests with different approval scenarios

4. Performance Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
PT-01	Load the Software Installation Request catalog form with all required variables.	Form should load within 2 seconds .	Form loaded successfully within the expected time.	Pass
PT-02	Submit a software installation request from an employee.	Request and approval record should be created within 5 seconds .	Request processed successfully within the expected time.	Pass
PT-03	Submit five software installation requests simultaneously using different employee accounts.	All requests should generate independent approval records without data conflicts.	All requests were processed correctly with no data overlap.	Pass
PT-04	Manager approves a request through the My Approvals workspace.	Request status should update and notification should be sent within 5 seconds .	Workflow completed successfully and notifications were generated.	Pass
PT-05	Place a request in the On Hold state to trigger the Business Rule.	An Incident should be created automatically within 3 seconds , with no duplicate incidents.	Incident was created successfully without duplication.	Pass

5. Performance Observations

The following observations were recorded during performance testing:

- The Service Catalog form loaded quickly and consistently for all test executions.
- The Flow Designer workflow executed successfully without delays or timeout errors.
- Approval records were created immediately after request submission.
- Email notifications were delivered successfully after approval and rejection actions.
- Business Rules executed efficiently and created Incidents only when the required conditions were met.
- Concurrent request submissions did not result in data inconsistencies or workflow failures.
- No noticeable performance degradation was observed during testing on the Personal Developer Instance.

6. Performance Test Summary

Performance Metric	Result
Catalog Form Loading	Within acceptable response time
Request Submission	Successful
Approval Workflow Execution	Successful
Catalog Task Generation	Successful
Incident Creation	Successful
Concurrent Request Handling	Successful
Email Notifications	Successful
Overall Performance	Satisfactory

7. Conclusion

The performance testing results demonstrate that the **Employee Software Installation Request** application performs reliably within the defined testing scope. The Service Catalog, Flow Designer workflows, Business Rules, approval process, and notification mechanisms operated efficiently without significant delays or failures.

Since the application was developed and tested on a **ServiceNow Personal Developer Instance (PDI)**, testing was limited to a moderate number of concurrent users suitable for an academic project. For deployment in an enterprise environment, additional large-scale load, stress, and scalability testing is recommended to validate system performance under production-level workloads.